

Intelligent event-triggering fault estimation and fault tolerant control of a quadrotor UAV with actuator fault

Zike Yuan, Xiaoxu Liu, Wenwei Zhang

Abstract—Quadrotor UAVs have a wide range of applications in production and daily life. To address the issues of observation and fault-tolerant control of UAVs in these applications, we propose a networked control-based event-triggered observation and fault-tolerant control scheme. In our research, by designing dynamic event-triggered thresholds, we can not only observe faults and system states in real time but also implement effective fault-tolerant control compensation based on the observation results. To dynamically optimize the event-triggering thresholds, we use the Proximal Policy Optimization (PPO) algorithm from reinforcement learning for training. By intelligently adjusting the triggering conditions, our scheme not only improves the UAV's robustness to motor faults but also saves computational resources by reducing unnecessary control updates. Through simulations and physical experiments, we have verified the effectiveness of the proposed method. The experimental results prove that the strategy significantly enhances the UAV's fault tolerance capabilities in the event of motor faults, ensuring the reliability and efficiency of UAVs when performing critical tasks. This research provides new technical support for the operation of UAVs in application environments.

Index Terms—UAV, event-triggered mechanism, RL, PPO algorithm, augmented system approach, fault estimation, fault tolerant control

I. INTRODUCTION

QUADROTOR unmanned aerial vehicles (UAVs) have become an essential tool across a variety of fields, thanks to their outstanding agility. These versatile machines are employed in a range of tasks from monitoring agricultural lands [1] to conducting search and rescue operations in disaster-stricken areas [2]. They also hold significant promise for applications in sectors such as electric power inspections [3] and transportation logistics [4]. Furthermore, quadcopters are extensively utilized in aerial photography, delivering high-resolution imagery for both cartographic and surveillance purposes.

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As quadcopters are increasingly deployed in multifaceted missions, their integration into networked control systems becomes more critical. Network control enables the real-time exchange of data between drones and ground control stations, enhancing situational awareness and improving operational command. Within this framework, event-triggered control (ETC) has emerged as a promising paradigm for orchestrating communication and control processes in networked UAVs. Unlike traditional methods based on fixed intervals, ETC strategies are designed to initiate actions in response to specific events or conditions that signal the necessity for system adjustments. Under ETC systems, signals are dispatched to the controller only when predefined trigger conditions are met [5]–[7]. This method offers the potential to reduce communication overhead and conserve computational resources, which is particularly beneficial for energy-limited devices like drones. To date, event-triggered techniques have garnered substantial achievements in the realm of networked control and monitoring challenges [8]–[11]. These methods have been successfully applied in various control systems, including energy network management and swarm coordination [12]–[14]. More recently, many studies have focused on integrating event-triggered control with reinforcement learning. For instance, [15] has proposed an innovative approach that combines event-triggered mechanisms with reinforcement learning to dynamically adjust the event-triggering threshold. This method effectively conserves resources while maintaining precision in observation and control. Event-triggered communication strategies based on intelligent algorithms have demonstrated significant effectiveness in various applications, including nonlinear system control [16], multi-agent systems [17], and intelligent driving control [18], achieving notable reductions in network communication overhead. However, the application of these methods to fault system observation and fault-tolerant control remains relatively underexplored, highlighting a promising direction for future research.

In recent years, the fields of fault diagnosis and fault-tolerant control have garnered significant interest and research from scholars, finding applications across various domains such as mobile robots, wind turbine benchmark system, aircraft engine, and some complex nonlinear systems [19]–[22]. The fault diagnosis and fault-tolerant control of UAVs represent a particular focus within this body of research. One study [23] explores a safety control method for quadcopters with damaged blades, offering a way to manage UAVs despite

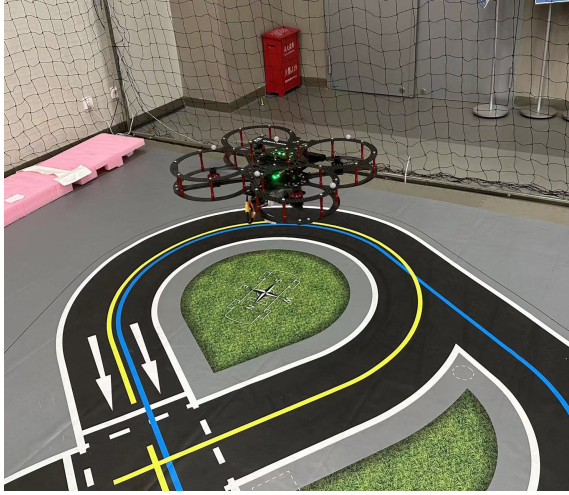


Fig. 1. The Qdrone quadcopter drone used during the experiment

blade impairments. Another piece of research [24] introduces a dual-loop nonlinear disturbance and fault observer method, which aids in trajectory planning during faults to enhance flight safety performance. Further, in [25], an investigation into nonlinear adaptive estimation techniques is presented for system adjustments in the presence of actuator faults. As computational capabilities of devices have advanced, learning-based approaches have seen progress in the area of fault-tolerant control. Works such as [26] and [27] have utilized a hybrid strategy that combines a baseline controller with a reinforcement learning controller. This approach leverages the auxiliary strategy of reinforcement learning to facilitate fault-tolerant control, ensuring the stable flight of quadcopters.

The synergy of event-triggered mechanisms with fault detection and fault tolerance represents a compelling area of research [28]. Given the limited communication and computational resources on UAVs, constant fault monitoring is not a resource-efficient approach. The adoption of an event-triggered mechanism reduces the need for periodic signal transmission, enabling communication only when trigger conditions are met. This, to some extent, decreases the system's reliance on communication resources. While existing studies, such as [29] and [15], have explored event-triggered control and fault detection individually, their integration into a unified framework remains underexplored. Our study bridges this gap by focusing on the fusion of event-triggered control with fault detection and fault-tolerant control strategies within a networked control system. This integration is intended to bolster the robustness and dependability of quadcopters, particularly in their expanding application spectrum.

The major contributions of this work are as follows:

- 1) **Event-Triggered State and Fault Estimation for a UAV:** We propose an event-triggered fault estimation framework, where state and fault estimation are integrated into a unified process. Inspired by [22], an augmented system is constructed by incorporating system states and fault signals into an extended state vector. Then, an event-triggered observer is developed to estimate states and faults simultaneously. Theoretical stability is ensured

through Lyapunov analysis. Compared with existing fault diagnosis methods for UAV, the presented technique can be applied to system with constrained communication and computational resources.

- 2) **Dynamic Threshold Adjustment Using Reinforcement Learning:** Unlike [11], which uses event-triggered control primarily to reduce communication burden, we introduce a reinforcement learning-based method to dynamically adjust triggering thresholds. This approach adapts thresholds based on system conditions, balancing communication efficiency and fault detection accuracy. Compared to fixed-threshold strategies, this method provides greater flexibility and precision. Also, different from adaptive or other dynamic event-triggered mechanisms, our proposed mechanism eliminates the need for designing specific triggering rules. By optimizing threshold triggering strategies through input-output signals and reward mechanisms, the framework can be effectively generalized to various observed control objects.
- 3) **Event-Triggered Fault tolerant control based on signal compensation, Practical Implementation:** Based on the estimated fault signals, we design a fault-tolerant control strategy through actuator signal compensation. Theoretical proof is provided to ensure system stability and robustness after fault tolerant control. Furthermore, the proposed method is validated through simulations and hardware experiments of a UAV, demonstrating improvements in fault tolerance and communication efficiency.

Our manuscript is structured as follows: Section II provides a detailed account of the UAV dynamics model along with the fault scenarios addressed by our method. Section III is dedicated to the construction of an augmented state-free UAV dynamics and human-machine state space model. In this section, we apply the event-triggered mechanism to detect failures within a quadrotor UAV that operates under a pre-existing PID controller. We also introduce a fault-tolerant compensation control strategy. To enhance the efficiency of the event-triggered mechanism, Section III further explores the use of a reinforcement learning-based approach to dynamically adjust the event-triggering threshold. Section IV validates the efficacy of our proposed method with a series of simulations and practical implementations on real-world platforms.

II. SYSTEM DESCRIPTION AND PROBLEM FORMULATION

A. UAV Model Description

The dynamic behavior of a quadrotor drone is characterized by its underactuated nonlinear system properties, with multiple inputs and outputs governed by four independent motors. Each motor's thrust is a function of the Pulse Width Modulation (PWM) signal it receives, and this relationship can typically be modeled by a linear equation:

$$T_i = K_i u_i \quad (1)$$

where T_i represents the thrust produced by the i -th motor, K_i is the thrust coefficient for that motor, and u_i is the PWM

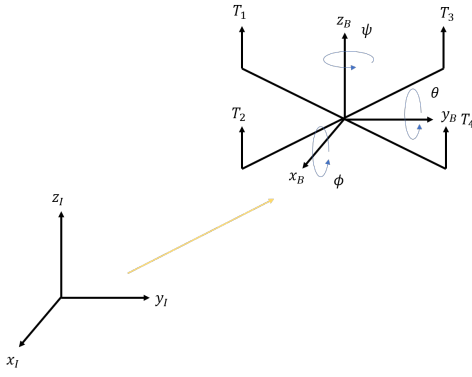


Fig. 2. Quadcopter coordinate system representation

input to the motor. The total thrust (T_m) of the drone can be calculated by summing the thrust from each motor:

$$T_m = T_1 + T_2 + T_3 + T_4. \quad (2)$$

As detailed in [30], the dynamic model of a quadrotor can be described in terms of the body frame B and the inertial frame I . These coordinate systems and their relationship are illustrated in Figure 2. Euler angles are defined as $\Omega = [\phi, \theta, \psi]^T$, representing roll, pitch, and yaw. The transformation matrix that relates the body frame to the inertial frame, ${}^I R_B$, is given by:

$${}^I R_B = \begin{bmatrix} c_\psi c_\theta & -s_\psi c_\theta + c_\psi s_\theta s_\phi & s_\psi s_\theta + c_\psi s_\theta c_\phi \\ s_\psi c_\theta & c_\psi c_\theta + s_\psi s_\theta s_\phi & -c_\psi s_\theta + s_\psi s_\theta c_\phi \\ -s_\theta & c_\theta s_\phi & c_\theta c_\phi \end{bmatrix}. \quad (3)$$

Based on the translational and rotational dynamics of the drone, we can derive the following dynamical system for the drone

$$\begin{cases} m\ddot{\mathbf{p}}_e + mg\mathbf{e}_3 = \mathbf{F} + \mathbf{F}_d \\ \mathbf{J}\dot{\boldsymbol{\omega}}_b + \boldsymbol{\omega}_b \times (\mathbf{J}\boldsymbol{\omega}_b) = \boldsymbol{\tau} + \boldsymbol{\tau}_d \end{cases} \quad (4)$$

where $\mathbf{e}_3 = [0, 0, 1]^T$, m, g represent the total mass of the UAV and the acceleration due to gravity, respectively. $\mathbf{p}_e = [x, y, z]^T$, $\mathbf{F} = {}^I R_B T_m \mathbf{e}_3 = [F_x, F_y, F_z]^T$, $\boldsymbol{\tau} = [\tau_x, \tau_y, \tau_z]^T$ specify the position, force, and torque, respectively. $\mathbf{F}_d$ and $\boldsymbol{\tau}_d$ represent the disturbances acting on the forces and torques. $\mathbf{J} = \text{diag}[J_{xx}, J_{yy}, J_{zz}]$ represent the inertia moment relative to the center of mass. $\boldsymbol{\omega}_b = [p, q, r]^T$ is the rotation angular rate in the body coordinate system. The relationship between $\boldsymbol{\omega}_b$ and the Euler angular rate $\dot{\Omega}$ can be obtained by rotating the matrix $\mathbf{T}$. It can be described as $\dot{\Omega} = \mathbf{T}\boldsymbol{\omega}_b$, where $\mathbf{T}$ can be expressed as

$$\mathbf{T} = \begin{bmatrix} 1 & s_\phi t_\theta & c_\phi t_\theta \\ 0 & c_\phi & -s_\phi \\ 0 & s_\phi / c_\theta & c_\phi / c_\theta \end{bmatrix}. \quad (5)$$

The mapping relationship among the generated thrust, torque, and PWM inputs can be simplified as

$$\begin{bmatrix} T_m \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} K_f & K_f & K_f & K_f \\ -d_\theta K_f & -d_\theta K_f & d_\theta K_f & d_\theta K_f \\ d_\phi K_f & -d_\phi K_f & d_\phi K_f & -d_\phi K_f \\ K_t & K_t & K_t & K_t \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \\ u_4 \end{bmatrix} \quad (6)$$

The $[u_1, u_2, u_3, u_4]^T$ represent the PWM input for each motors. d_θ and d_ϕ refer to the half distance of roll and pitch rotor to rotor, respectively. K_f and K_t represent the coefficients of thrust and torque, respectively. Consider the flight of a quadcopter at a small angle. It is assumed that the roll motion and pitch motion are so slight that $\sin\theta = \theta$ and $\sin\phi = \phi$, and no yaw motion generate throughout the flight ($\psi = 0$). Combined with (2), the following UAV dynamics model can be constructed as

$$\begin{cases} \ddot{x} = \theta g + \omega_x \\ \ddot{y} = -\phi g + \omega_y \\ \ddot{z} = \frac{T_m}{M} - g + \omega_z \\ \ddot{\theta} = \frac{u_y}{J_{yy}} + \omega_\theta \\ \ddot{\phi} = \frac{u_x}{J_{xx}} + \omega_\phi \\ \ddot{\psi} = \frac{u_z}{J_{zz}} + \omega_\psi \end{cases} \quad (7)$$

Define $x = [x \ \dot{x} \ y \ \dot{y} \ \int z dt \ z \ \dot{z} \ \theta \ \dot{\theta} \ \phi \ \dot{\phi} \ \psi \ \dot{\psi}]^T$ as the system state vector, consists of the position, attitude and speed information of the Drone. $\omega = [\omega_x \ \omega_y \ \omega_z \ \omega_\theta \ \omega_\phi \ \omega_\psi]^T$ as the external disturbance vector. $u = [u_T \ u_y \ u_x \ u_z]^T$ as the control input vector. $y = [x \ y \ z \ \theta \ \phi \ \psi]^T$ as the output vector. The state space equation obtained from dynamic model (7) can be expressed as

$$\begin{cases} \dot{x}(t) = Ax(t) + Bu(t) + D\omega(t) + Gg \\ y(t) = Cx(t) \end{cases} \quad (8)$$

with $x \in \mathbb{R}^{13}$, $u \in \mathbb{R}^4$, $\omega \in \mathbb{R}^6$ and $y \in \mathbb{R}^6$. $A \in \mathbb{R}^{13 \times 13}$ being the quadrotor's system matrix, $B \in \mathbb{R}^{13 \times 4}$ being the input matrix. $D \in \mathbb{R}^{13 \times 6}$ is the disturbance matrix, which describes the influence of random disturbances, $G \in \mathbb{R}^{13}$ is a vector related to gravity, and g is the scalar gravitational acceleration. $C \in \mathbb{R}^{6 \times 13}$ represents the output matrix. During the flight process, there exist random system noise from the environment and random measurement noise, which are uniformly modeled as a high-frequency small perturbation. The perturbation ω in simulation satisfies $\omega \sim \mathcal{N}(\mathbf{0}, \mathbf{Q})$, where $\mathbf{Q} = \text{diag}\{0.01, 0.01, 0.01, 0.01, 0.01, 0.01\}$.

Considering using a PID controller to control the UAV, the control input can be designed as

$$\begin{aligned} u_T &= K_z \begin{bmatrix} \int z dt \\ z \\ \dot{z} \end{bmatrix} + mg \\ u_x &= K_x \begin{bmatrix} x \\ \dot{x} \end{bmatrix} + K_\phi \begin{bmatrix} \phi \\ \dot{\phi} \end{bmatrix} \\ u_y &= K_y \begin{bmatrix} y \\ \dot{y} \end{bmatrix} + K_\theta \begin{bmatrix} \theta \\ \dot{\theta} \end{bmatrix} \\ u_z &= K_\psi \begin{bmatrix} \psi \\ \dot{\psi} \end{bmatrix} \end{aligned} \quad (9)$$

B. Formulation Of Actuator Faults

In this paper, we focus on actuator failures commonly encountered by quadcopters, particularly those affecting the motors and propellers. One frequent cause of such failures is propeller damage, which leads to reduced lift. Additionally,

motor failures may arise due to overloading, bearing degradation, or thermal issues, all of which can contribute to a slower rotor speed. These degradations not only diminish performance but also pose significant risks to the reliability and safety of UAVs during operation.

The actuator fault can be mathematically described as:

$$w_i^f = u_i - \Delta_i \quad (10)$$

where u_i^f denotes the control input of the i -th motor under fault conditions, and Δ_i represents the magnitude of the bias fault affecting the i -th motor. Specifically, Δ_i can take different forms depending on the nature of the fault. For instance, if the fault is caused by a loss of lift due to propeller damage or a slower rotor speed, Δ_i can be modeled as $\Delta_i = (1 - \gamma)u_i$, where $\gamma \in (0, 1)$ represents the degradation factor of the motor. If the fault is caused by persistent external disturbances or mechanical wear, Δ_i can be represented as a constant bias: $\Delta_i = F$, where F accounts for the additional force or resistance offset introduced by the fault.

III. EVENT-TRIGGERED AUGMENTED SYSTEM FAULT OBSERVATION AND TOLERANT CONTROL BASED ON REINFORCEMENT LEARNING

A. System Description and Augmented System

Considering the actuator fault, the system (8) can be described by the following state-space equations

$$\begin{cases} \dot{x} = Ax(t) + Bu(t) + D\omega(t) + B_f f(t) + Gg \\ y(t) = Cx(t) \end{cases} \quad (11)$$

where $f \in \mathbb{R}^4$ indicates the faults occurring on each actuator, $B_f \in \mathbb{R}^{13 \times 4}$ represents the fault matrix.

Taking into account the failure of the motor, it will not fluctuate violently. Let the $\dot{f} = 0$ and define $\bar{x} = \begin{bmatrix} x \\ f \end{bmatrix}$. We can construct an equivalent augmented system as follows:

$$\begin{cases} \dot{\bar{x}}(t) = \bar{A}\bar{x}(t) + \bar{B}u(t) + \bar{D}\omega(t) + \bar{G}g \\ y(t) = \bar{C}\bar{x}(t) \end{cases} \quad (12)$$

where $\bar{x} \in \mathbb{R}^{21}$, $\bar{A} \in \mathbb{R}^{21 \times 21}$, $\bar{B} \in \mathbb{R}^{21 \times 4}$, $\bar{D} \in \mathbb{R}^{21 \times 6}$, $\bar{G} \in \mathbb{R}^{21}$, $\bar{C} \in \mathbb{R}^{6 \times 21}$ and can be express as

$$\bar{A} = \begin{bmatrix} A & 0 & B_f \\ 0 & 0 & 0 \\ 0 & I & 0 \end{bmatrix}, \bar{B} = \begin{bmatrix} B \\ 0 \\ 0 \end{bmatrix}, \bar{C} = [C \quad 0 \quad 0]$$

$$\bar{D} = \begin{bmatrix} D \\ 0 \\ 0 \end{bmatrix}, \bar{G} = \begin{bmatrix} G \\ 0 \\ 0 \end{bmatrix}$$

Remark 3.1: For an augmented system such as (12), an observer can be constructed to enable the simultaneous observation of both system states and faults. Specifically, the augmented system integrates the fault dynamics into the overall system model, allowing the observer to estimate faults as part of the state vector. This approach ensures that the observation of faults and states is unified within the same framework.

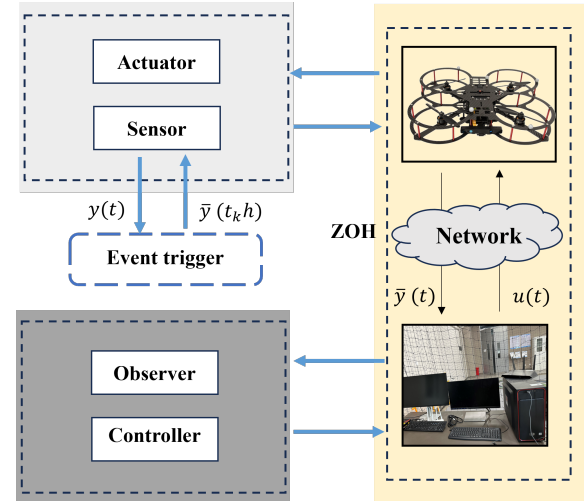


Fig. 3. Event-Triggered fault estimation and fault-tolerant control

B. Intelligent Event-Triggered Fault Estimate and Fault Tolerance Control

Due to the fact that the limit of communication resource and bandwidth of network. Communication in every instant is unrealistic. In this section, we design a dynamic event-triggered communication Mechanism, Which used RL algorithm to dynamically adjust event-triggered thresholds. The triggering only occurs when the following condition is satisfied:

$$\theta^T(t)M\theta(t) \geq \delta(j_k h)y^T(j_k h)My(j_k h) \quad (13)$$

where $t_k h$ and $j_k h$ represent the trigger time and sampling time respectively, $\theta(t) = y(t_k h) - y(j_k h)$, $\delta(j_k h)$ represents the dynamic triggering threshold, which will be learned using the RL algorithm. It satisfies the condition $\delta \leq \delta_{\max}$, where $\delta_{\max}$ denotes the maximum triggering threshold. $M > 0$ is the weighting parameter which will be designed latter.

In the study of network-based control systems, the frequency of information transmission is significantly influenced by the threshold value δ . A lower threshold often leads to an increased rate of communication, as the system states are more likely to exceed this smaller threshold, thereby triggering corresponding events and control responses. This can enhance the performance of the control system but also increases the network load and energy consumption. In the event-triggered fault detection and tolerance mechanism proposed in this research, we have designed an event-triggered control strategy based on reinforcement learning. Against the backdrop of a pre-established controller performing flight control tasks, our developed observer utilizes an event-triggered mechanism to monitor the system states in real time and identify faults. Utilizing reinforcement learning techniques, it dynamically adjusts the threshold values. In instances where a fault occurs leading to system fluctuations, the strategy selects a lower communication threshold to ensure accurate monitoring of the system and the fault, thus ensuring precise and rapid fault compensation. Conversely, when the system maintains a stable and healthy state, it automatically increases the communication

threshold, thereby reducing the demand for communication and computation and effectively saving resources.

To achieve a dynamic triggering threshold output, this paper presents a policy-based reinforcement learning algorithm. This algorithm is framed as a Markov decision process (MDP), which enables the translation of dynamic event-triggered control problems into a Markov model paradigm. The model is encapsulated by the tuple $(\mathbf{S}, \mathbf{A}, \mathbf{P}, \gamma, r)$, where $\mathbf{S}$ denotes the state space, $\mathbf{A}$ represents the action space, and $\mathbf{P}$ is the state transition probability function. The state $\mathbf{S}$ is defined by $s = [y_n - y_{n-1} \quad \delta_n]^\top$. The action space $\mathbf{A}$ consists of all feasible actions that the agent can take, with an action a defined as $a = \delta(j_k h)$, indicating that the agent's output should serve as the threshold for the event-triggered condition. The transition function $\mathbf{P}$ characterizes the probabilities of transitioning to the next state s_{k+1} after taking an action a_k in the current state s_k , and is formally defined as $\mathbf{P} : \mathbf{S} \times \mathbf{A} \rightarrow \Delta(\mathbf{S})$, where $\Delta(\mathbf{S})$ is the set of probability distributions over the state space $\mathbf{S}$. The discount parameter $\gamma \in [0, 1]$ specifies the discount factor that assures the convergence of the expected sum of rewards. The reward function r evaluates the desirability of the agent's action a in the context of the current state s . The goal is to infer an optimized dynamic triggering threshold policy that minimizes the frequency of triggers while preserving system stability. The reward function r is designed as follows:

$$r_k = \sum_{i=1}^n \frac{K_i |y_i(k) - \hat{y}_i(k)|}{|y_i(k)|} + K_{n+1} (\delta_k - 0.1)^2. \quad (14)$$

Our reward function can be understood as consisting of two parts. In the first part the $y_i(k)$ represents the true observation value at the current time step k for the i -th step (used only during the training process), and $\hat{y}_i(k)$ represents the observed value obtained through the event-triggered observer. n represents the ratio of the threshold selection period to the system sampling period. This part of the reward function encourages the event-triggered observer to choose triggering thresholds that allow the observed values to approximate the true values as closely as possible (in general, more frequent triggering leads to values closer to the true ones). The second part of the reward function introduces a penalty term designed to encourage the selection of larger thresholds. By combining these two components, the system is guided to comprehensively consider the selection of thresholds based on the current state.

To solve the optimization problem defined above, we employ the Proximal Policy Optimization (PPO) algorithm [31]. PPO is chosen for its stability and efficiency in policy updates, achieved by limiting the divergence between the new policy and the old policy through a clipped surrogate objective. The policy network outputs actions (i.e., the triggering thresholds), while the value network estimates the expected cumulative rewards to guide learning.

The agent's policy $\pi_\theta(a|s)$ is parameterized by a neural network with parameters θ , and optimized using the PPO loss function:

$$L^{\text{PPO}}(\theta) = \mathbb{E} \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1 - \epsilon, 1 + \epsilon) \hat{A}_t \right) \right] \quad (15)$$

where $r_t(\theta) = \frac{\pi_\theta(a_t|s_t)}{\pi_{\theta^{\text{old}}}(a_t|s_t)}$ is the probability ratio between the new and old policies, $\hat{A}_t$ is the advantage function approximated using Generalized Advantage Estimation (GAE), and ϵ is the clipping parameter that controls the policy update step size.

During training, the system state s is fed as input to the policy network, which outputs the optimal triggering threshold a . The reward function r_k guides the learning process, ensuring that the selected thresholds balance the trade-off between communication efficiency and system stability. The PPO algorithm iteratively updates the network parameters until convergence, resulting in a policy that adapts to varying system states and optimally adjusts the triggering thresholds.

Within this group, the former serves as a reward to ensure state convergence, while the latter acts as an incentive to minimize the number of event triggers. Moreover, let $\forall a_k \in \mathbf{A}$, $a_k \leq \delta_{\max}$, the following inequality can be obtained from (11)

$$\theta^T(t) M \theta(t) \geq \delta_{\max} y^T(j_k h) M y(j_k h). \quad (16)$$

Taking the behavior of zero-order-holder (ZOH) into consideration, the input signal of system output can be represented by $\bar{y}(t) = y(t_k h)$, and following fault estimation observer is constructed:

$$\begin{cases} \dot{\hat{x}}(t) = \bar{A} \hat{x}(t) + \bar{B} u(t) + L(\bar{y}(t) - \hat{y}(t)) + \bar{G} g \\ y(t) = \bar{C} \hat{x}(t) \end{cases} \quad (17)$$

After deriving the estimated value of the fault, it becomes feasible to devise a fault compensation mechanism that facilitates fault-tolerant control. The control input, denoted by u , is formulated as $u = Kx - Gmg$, with K representing the gain parameter associated with the PID controller. Consequently, the dynamical representation of the system, incorporating the fault compensation strategy, is expressed as follows:

$$\dot{x}(t) = (A + BK)x(t) + D\omega(t) + B_f(f(t) - \hat{f}(t)). \quad (18)$$

Remark 3.2, [32]: We note that the event-triggered mechanism proposed in (13) falls under the category of periodic event-triggered control [33]. Additionally, the minimum inter-event time is guaranteed to be at least one sampling period, thereby inherently satisfying the event-separation properties outlined in [34], [35]. Therefore, the Zeno phenomenon does not occur.

Define $\xi = [e(t) \ x(t)]^T = [e_x(t) \ \dot{e}_f(t) \ e_f(t) \ x(t)]^T$, where $e_x(t) = \hat{x}(t) - x(t)$ represents the difference between the estimated state and the actual state, and $e_f(t) = \hat{f} - f$ represents the difference between the estimated error and the actual error. The system can be written as

$$\dot{\xi}(t) = \mathbb{A}\xi(t) + \mathbb{L}\theta(t) + \mathbb{D}\omega(t) \quad (19)$$

where

$$\mathbb{A} = \begin{bmatrix} \bar{A} - L\bar{C} & 0 \\ \bar{B}_f & A + BK \end{bmatrix}, \mathbb{L} = \begin{bmatrix} L \\ 0 \end{bmatrix}, \mathbb{D} = \begin{bmatrix} \bar{D} \\ D \end{bmatrix}$$

$$\bar{B}_f = [0 \quad 0 \quad B_f]$$

Theorem 1 : For given scalars δ, γ , there exists L that makes the observer system stable $\mathbb{P} > 0, M > 0$ such that

$$\Pi = \begin{bmatrix} \Pi_{11} & \Pi_{12} & \Pi_{13} & \Pi_{14} \\ * & \Pi_{22} & 0 & \Pi_{24} \\ * & * & -\Omega & 0 \\ * & * & * & -\gamma^2 \end{bmatrix} < 0 \quad (20)$$

where

$$\begin{aligned} \Pi_{11} &= \mathbf{H}e\{P_1\bar{A} - P_1L\bar{C}\} \\ \Pi_{12} &= \bar{B}_f^T P_2 \\ \Pi_{13} &= P_1 L \\ \Pi_{14} &= P_1 \bar{D} \\ \Pi_{22} &= \mathbf{H}e\{P_2 A + P_2 B K\} + \delta C^T M C + I \\ \Pi_{24} &= P_2 D \end{aligned}$$

$$\mathbb{P} = \begin{bmatrix} P_1 & 0 \\ 0 & P_2 \end{bmatrix}$$

If the above matrix inequality holds, and satisfy the H_∞ condition:

$$\|x\|_{T_f} \leq \gamma \|\omega\|_{T_f} \quad (21)$$

it can be shown that system (17) is stable.

Proof. For the Lyapunov functional

$$V(t) = \xi^T(t) P \xi(t) \quad (22)$$

We can get

$$\begin{aligned} \dot{V}(t) &= \xi^T(t) \mathbb{P} \mathbb{A} + \mathbb{A}^T \mathbb{P} \xi(t) + 2\xi^T(t) \mathbb{P} L \theta(t) \\ &\quad + 2\xi^T(t) \mathbb{P} D \omega(t). \end{aligned} \quad (23)$$

According to (13), the (23) can be written as

$$\begin{aligned} \dot{V}(t) &\leq \xi^T(t) \mathbb{P} \mathbb{A} + \mathbb{A}^T \mathbb{P} \xi(t) + 2\xi^T(t) \mathbb{P} L \theta(t) \\ &\quad + 2\xi^T(t) \mathbb{P} D \omega(t) + \delta x^T(t) C^T M C x(t) \\ &\quad - \theta^T(t) M \theta(t). \end{aligned} \quad (24)$$

Apparently, when $\omega(t) = 0$, one can obtain $\dot{V}(t) \leq 0$, implying that the system (19) is asymptotically stable.

For the purpose of establishing H_∞ performance of the system, construct function $J(t)$ as follows:

$$\mathcal{J}(t) = \dot{V}(t) + x^T(t)x(t) - \gamma^2 \omega^T(t)\omega(t). \quad (25)$$

It follows that

$$\begin{aligned} \mathcal{J}(t) &= \xi^T(t) \mathbb{P} \mathbb{A} + \mathbb{A}^T \mathbb{P} \xi(t) + 2\xi^T(t) \mathbb{P} L \theta(t) \\ &\quad + x^T(t)x(t) - \gamma^2 \omega^T(t)\omega(t) + 2\xi^T(t) \mathbb{P} D \omega(t) \\ &\leq \xi^T(t) \mathbb{P} \mathbb{A} + \mathbb{A}^T \mathbb{P} \xi(t) + 2\xi^T(t) \mathbb{P} L \theta(t) \\ &\quad + x^T(t)x(t) - \gamma^2 \omega^T(t)\omega(t) + 2\xi^T(t) \mathbb{P} D \omega(t) \\ &\quad + \delta x^T(t) C^T M C x(t) - \theta^T(t) M \theta(t) \\ &= \eta^T(t) \Pi \eta(t) \end{aligned} \quad (26)$$

where $\eta(t) = [e_x(t) \ x(t) \ \omega(t)]^T$

Then, it holds from (20) that $\mathcal{J}(t) \leq 0$. By intergrate the inequality in (25) from 0 to T_f , one can obtain

$$\begin{aligned} &\int_0^{T_f} x^T(t)x(t) - \gamma^2 \omega^T(t)\omega(t) + \dot{V}(t) dt \\ &= V(T_f) - V(0) + \int_0^{T_f} x^T(t)x(t) - \gamma^2 \omega^T(t)\omega(t) dt \\ &< 0 \end{aligned} \quad (27)$$

Since $\Pi \leq 0$ in terms of (13), and form (27), We satisfy the performance index of (20), and this completes the proof.

IV. SIMULATION AND EXPERIMENTAL RESULTS

In this section, we verify the effectiveness of our method in simulations and real experiments. The parameters of the drone are provided by the manufacturer. The parameters of the PID controller are as follows:

$$\begin{aligned} K_x &= [-3.53 \ -1.76] \\ K_y &= [-2.91 \ -1.45] \\ K_z &= [-6 \ -35 \ -28] \\ K_\theta &= [-2.25 \ -0.20] \\ K_\phi &= [-1.85 \ -0.16] \\ K_\psi &= [-0.59 \ -0.158] \end{aligned} \quad (28)$$

Based on the dynamics model of the quadrotor UAV and the fault model, we have constructed a simulation environment for fault observation and fault-tolerant control. We utilized Quanser's QDrone for both simulation and physical experiments. The specific parameters of the QDrone are shown in Table I. For the design of the training environment, we introduced a fault to one of the UAV's motors at random intervals during the simulation; the fault coefficient γ , was a random value between 0 and 0.3. Additionally, random noise was added to the pose data of the UAV within the simulation. In order to implement the PPO algorithm, the actor and critic network are both established by four layers, one feature input layer and three fully connected layers. In this study, the PPO algorithm was employed to train an agent with a discrete action space comprising ten actions, defined as $\{0.01, 0.02, \dots, 0.1\}$. Each training episode lasted 40s, equivalent to 200 time steps at an interval of 0.2s. The specific hyperparameters used during the training process are shown in Table II.

The training results are shown in the Fig. 4 After 2,000 training iterations, we achieved satisfactory results. The selection of dynamic event-triggered thresholds after the training process is illustrated in Fig. 5. A fault was injected into the simulation between 20s to 30s. It can be observed that during the fluctuations in the system state caused by the fault, our strategy tends to choose smaller event-triggered thresholds to ensure a more precise estimation of the system state and the fault. When the system is in a relatively stable state, the strategy we trained opts for larger triggering thresholds to reduce the computational cost of system communication resources.

We implemented the trained event-triggered model on the actual hardware. In the real QDrone, we induced a fault in motor number 1 during the 20-30s timeframe of the flight. Event-triggered sampling period $h = 0.02s$. We conducted experiments using the RL method, and the results for the system's

TABLE I
PARAMETERS OF THE QUADROTOR UAV

Parameter	Value
m	1.121 kg
d_θ	0.1758 m
d_ϕ	0.2136 m
J_{xx}	0.01 kgm ²
J_{yy}	0.0082 kgm ²
J_{zz}	0.0148 kgm ²

TABLE II
HYPERPARAMETERS FOR PPO

Hyperparameter	Value
Experience Horizon	600
Clip Factor	0.02
Entropy Loss Weight	0.01
Mini-Batch Size	128
Number of Epochs	3
GAE Factor	0.95
Discount Factor	0.95

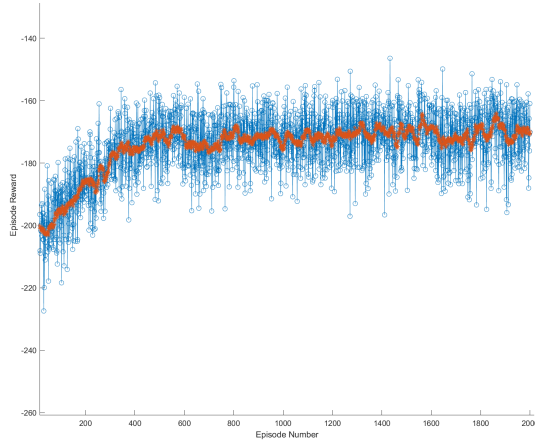


Fig. 4. Training Reward Curve

trigger and fault observation are shown in the corresponding figure. Based on the fault observed, we applied fault compensation using (13), as depicted in Fig. 6. We compared the states x of the healthy system, the faulty system, and the system after fault compensation. It can be observed that the state of the system after fault compensation is very close to that of the healthy system, which indicates the effectiveness of our fault-tolerant control method. The comparison between the estimated values of fault estimation and state estimation and the actual values is shown in Fig. 7 to Fig. 11. The observation of the state is the error between the actual state and the given state. The results indicate that we can achieve relatively accurate

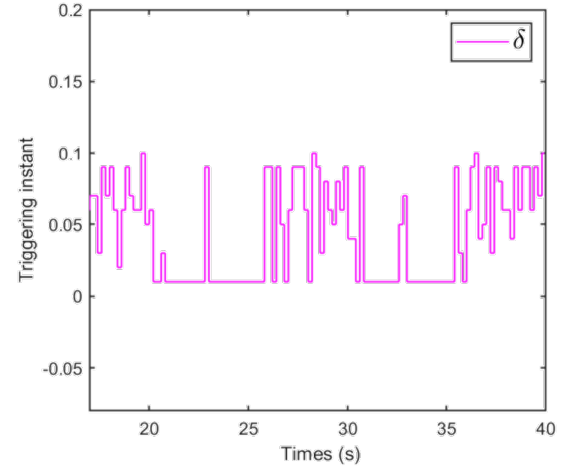


Fig. 5. Threshold selection of strategies in simulation

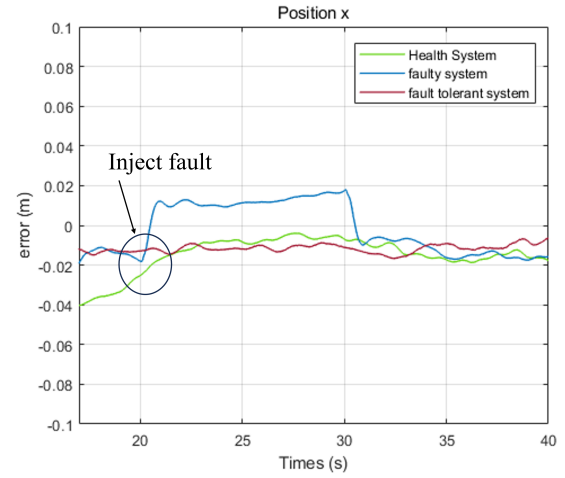


Fig. 6. State x under healthy system, faulty system and fault compensation system

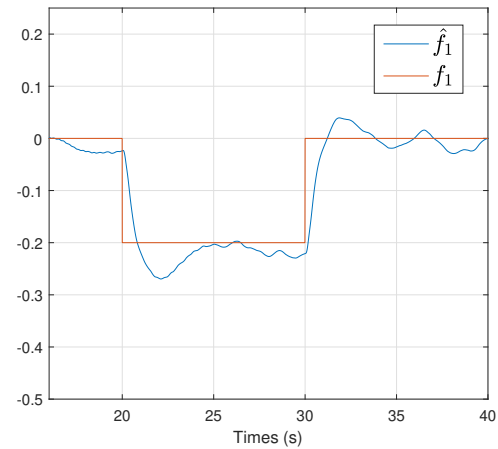


Fig. 7. Fault estimation in real flight

estimations for both the faults and the status.

In the practical implementation, the selection of thresholds

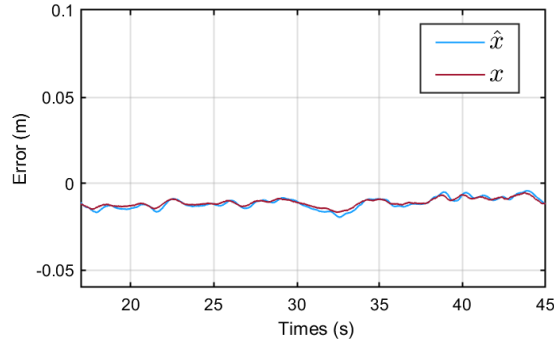


Fig. 8. State x estimation under real flight

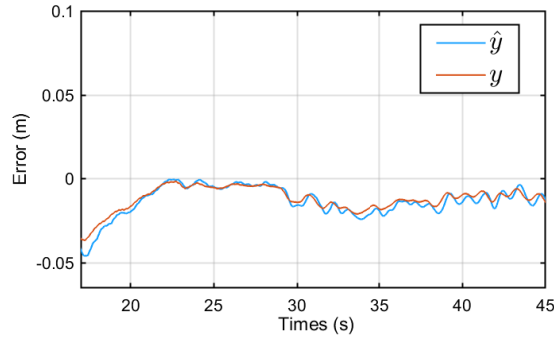


Fig. 9. State y estimation under real flight

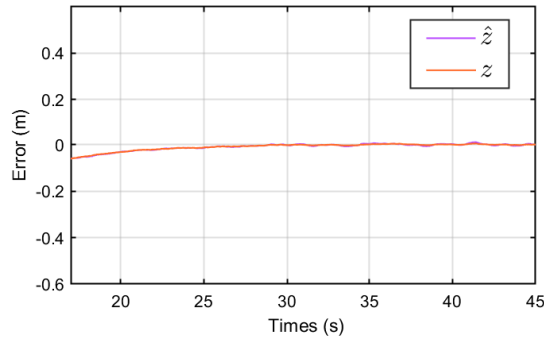


Fig. 10. State z estimation under real flight

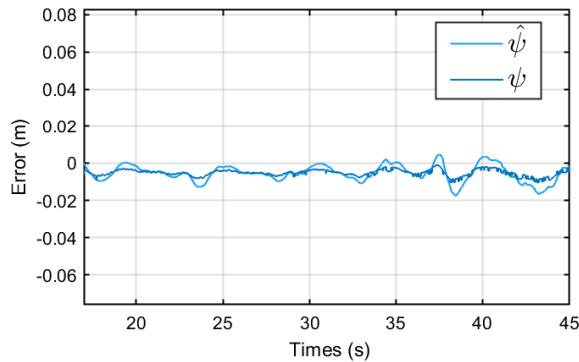


Fig. 11. Angle ψ estimation under real flight

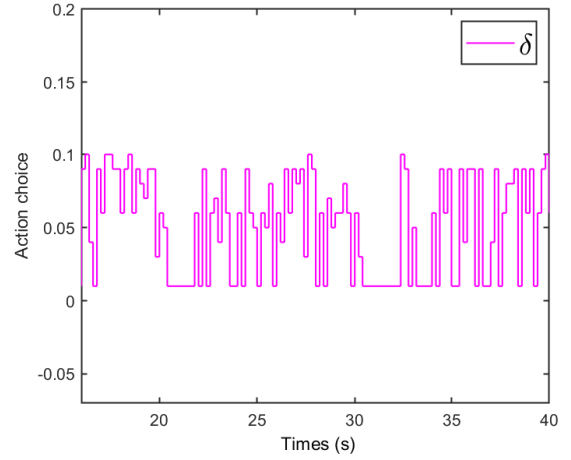


Fig. 12. Trigger threshold strategy during real flight

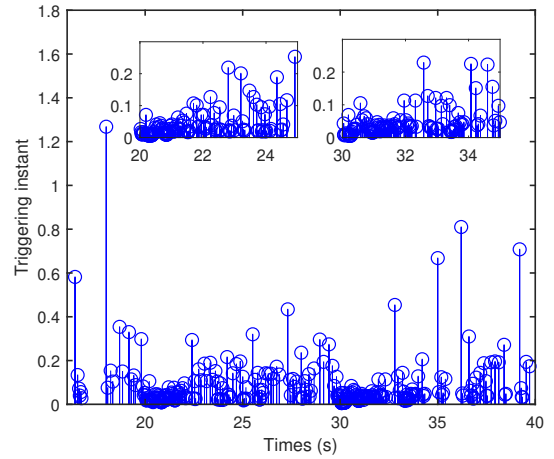


Fig. 13. Trigger interval for intelligent event triggering

for our strategy is depicted in Fig. 12. Concurrently, we evaluated our strategy against a fixed threshold event-triggered scheme. Physical tests were executed utilizing thresholds $\delta = 0.02$, $\delta = 0.05$, $\delta = 0.08$ for the fixed threshold event-triggered scheme. The interval between triggers for both the reinforcement learning (RL) based method and the fixed threshold strategy is presented in Fig. 13 - Fig. 16.

TABLE III
COMPARISON BETWEEN CONSTANT EVENT TRIGGERING AND RL

δ	0.02	0.05	0.08	RL
Trigger Times	1011	796	601	682
Max Error of state x ($\times 10^{-3}$ m)	1.41	2.04	2.53	0.98

We compare the state x using our proposed method with the constant threshold triggering method to highlight the advantages of our approach in Fig. 17. Specifically, we conduct comparisons at $\delta = 0.02$ and $\delta = 0.08$. It can be observed that our method achieves state observations closer to the system's true state, comparable to those obtained with a low triggering threshold. Meanwhile, the number of event triggers in our

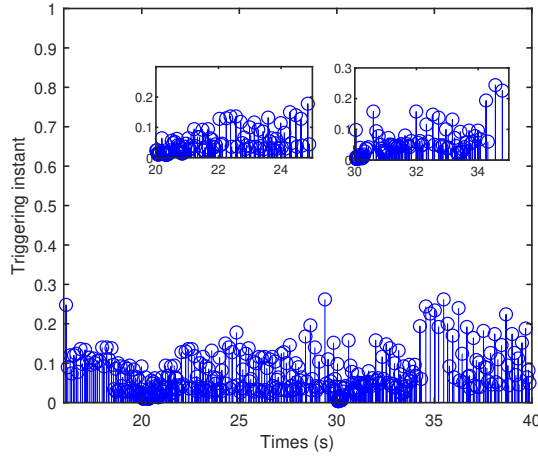


Fig. 14. Trigger interval for $\delta = 0.02$

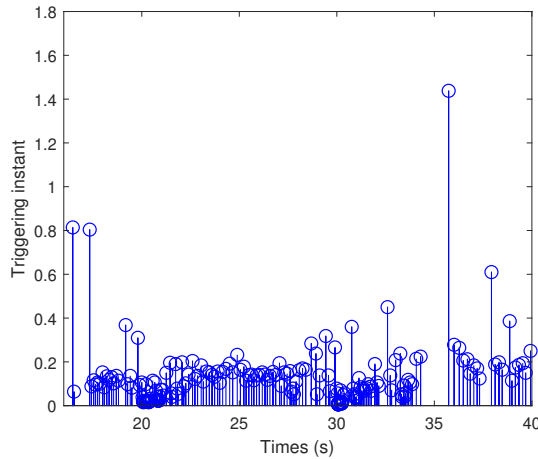


Fig. 15. Trigger interval for $\delta = 0.05$

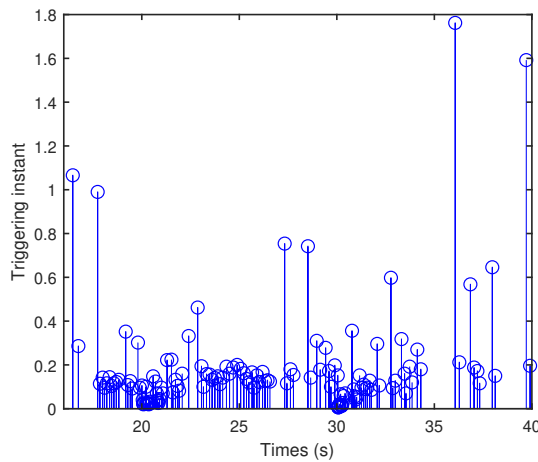


Fig. 16. Trigger interval for $\delta = 0.08$

method is significantly lower than that with the low threshold. This demonstrates that our approach effectively balances accurate state observation with the conservation of communication

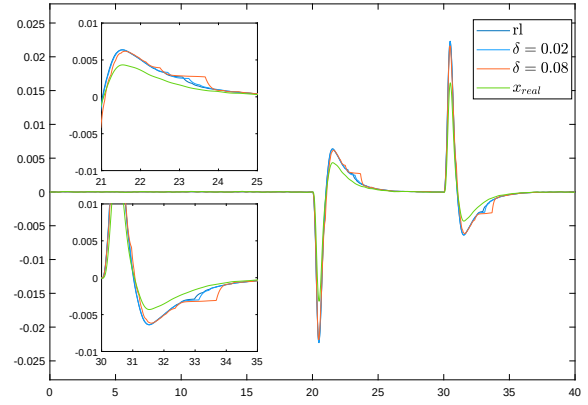


Fig. 17. Observation of state x under different event triggering methods

resources. Quantitative comparison of different methods is shown in Table III.

When contrasted with the fixed threshold triggering approach employing lower thresholds, our adaptive time-triggering strategy markedly diminished the trigger count in scenarios where the system was either faultless or in a state of equilibrium, thereby curtailing communication overhead. In scenarios where higher threshold values were compared, and the system underwent fault-induced oscillations, our strategy adeptly shifted to smaller thresholds to facilitate more frequent triggering. This resulted in a more precise detection of faults and assessment of the system's condition, enhancing the quality of fault remediation and state evaluation.

Our dynamic method is tailored to meet the system's varying requirements, ensuring communication efficiency during typical operations and heightened alertness in the event of faults. This presents a distinct benefit over conventional static triggering mechanisms that lack adaptability relative to the state of the system.

V. CONCLUSION

In this study, an intelligent event-triggered fault detection and fault-tolerant compensation control method for quadrotor UAVs was proposed. By leveraging networked fault detection and control strategies, the method effectively mitigates the effects of actuator failures, ensuring precise event-triggered observations and conserving communication resources. The stability of the system is guaranteed using Lyapunov's criterion, and the Proximal Policy Optimization (PPO) algorithm was employed for training. The proposed approach was successfully validated through deployment on an actual Qdrone. Future research will consider extending the method to address more complex nonlinear models and disturbances to enhance its applicability and robustness.

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